

Calc I Lecture 17: Mean Value Theorem

"mean" = "average"

Motivating example: You are driving along a highway,

speed limit = 60 mph

location at 2:15pm is 20 miles east of location at 2:00pm.

Can we conclude you went over the speed limit? (at any point)

$$\text{average velocity} = \frac{\text{displacement}}{\text{time}} = \frac{20 \text{ mi}}{1/4 \text{ hr}} = 80 \text{ mph}$$

But Can we conclude that your instantaneous speed was over the limit at some point in time?

Answer: Yes!

The Mean Value Theorem (MVT): Let $a < b$ be real numbers.
Let f be a function satisfying:

1. $f(x)$ is continuous on $[a, b]$
2. $f(x)$ is differentiable on (a, b)

Then there is a number c such that $a < c < b$ and

$$f'(c) = \frac{f(b) - f(a)}{b - a} \left. \vphantom{\frac{f(b) - f(a)}{b - a}} \right\} \begin{array}{l} \text{average rate of change of } f \text{ over } [a, b] \\ \text{change in } y \text{ over change in } x \end{array}$$

Connection to above example:

If $f(t)$ represents position at time t ,

$$\frac{f(b) - f(a)}{b - a} \left. \vphantom{\frac{f(b) - f(a)}{b - a}} \right\} \begin{array}{l} \text{change in position} = \text{displacement} \\ \text{change in time} \end{array} \left. \vphantom{\frac{f(b) - f(a)}{b - a}} \right\} \begin{array}{l} \text{average} \\ \text{velocity} \end{array}$$

$f'(c)$ } instantaneous velocity.

$\Rightarrow$ at some point in time between $t=a$, $t=b$,

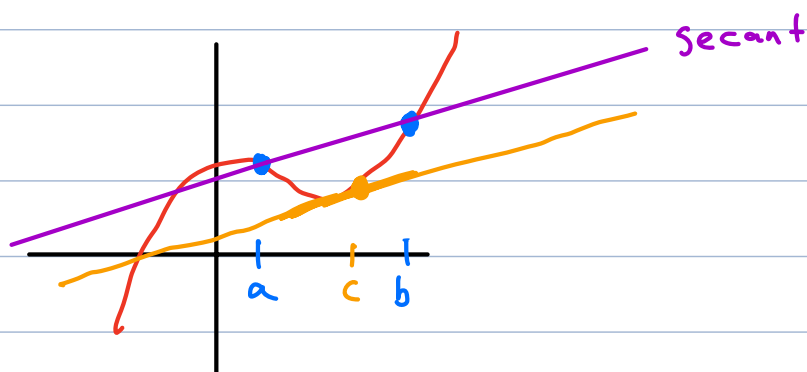
instantaneous velocity = average velocity over $[a, b]$

In particular, in above example, we can conclude that at some point in time, you were driving at 80 mph $\Rightarrow$ over speed limit

Graphical interpretation: let $y = f(x)$

$$\frac{f(b) - f(a)}{b - a} = \text{slope of } \underline{\text{secant}} \text{ line between } (a, f(a)) \text{ and } (b, f(b))$$

$$f'(c) = \text{slope of } \underline{\text{tangent}} \text{ line to } y = f(x) \text{ at } x = c.$$



(so for some c ,
the tangent line at c
parallel to secant line.)

Example Confirm that the conditions of MVT are satisfied for $f(x) = x^3$ over the interval $[0, 2]$

and find the value(s) of c that satisfy the conclusion of MVT over $[0, 2]$
 a b

Conditions:

1. $f(x)$ is continuous on $[a, b]$. Yes. x^3 continuous everywhere
2. $f(x)$ is differentiable on (a, b) Yes $f'(x) = 3x^2$ defined everywhere

$\Rightarrow$ conditions satisfied.

Value(s) of c that satisfy conclusion: $f'(c) = \frac{f(b) - f(a)}{b - a}$

$$\underbrace{f'(c)}_{3c^2} = \frac{f(2) - f(0)}{2 - 0} = \frac{2^3 - 0^3}{2} = \frac{8}{2} = 4 \Rightarrow 3c^2 = 4 \Rightarrow c^2 = \frac{4}{3}$$
$$c = \pm \frac{2}{\sqrt{3}}$$

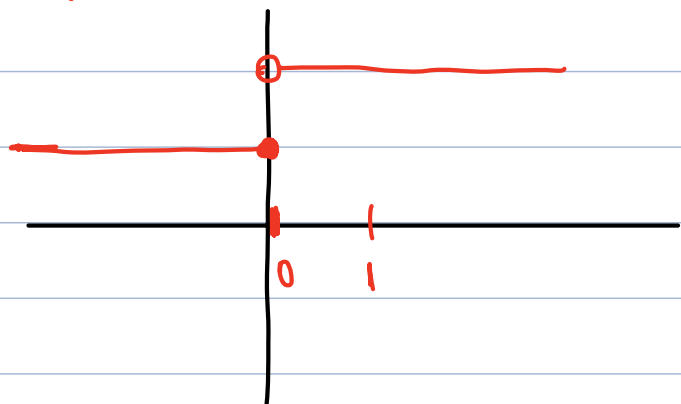
Non-examples

$$\Rightarrow \boxed{c = \frac{2}{\sqrt{3}}}$$

(need $a < c < b$)

If assumption 1) fails... (not continuous)

e.g.



$$f(x) = \begin{cases} 1 & \text{if } x \leq 0 \\ 2 & \text{if } x > 0 \end{cases}$$

$$f'(x) = 0 \text{ if } x \neq 0$$

undefined at $x = 0$

if we tried to apply MVT to $[a, b]$:

$$a < c < b$$

$$f'(c) \stackrel{?}{=} \frac{f(b) - f(a)}{b - a}$$

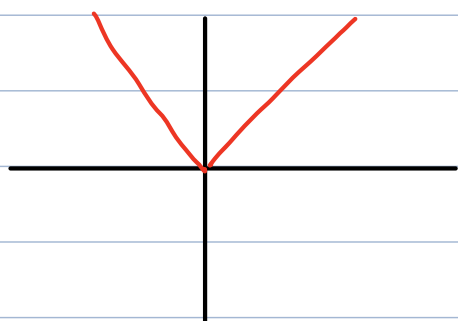
$\uparrow$ $\uparrow$
 0 $\frac{f(1) - f(0)}{1 - 0} = 1$

$0 \neq 1$. MVT fails

If assumption 2) fails...

(not differentiable)

e.g. $f(x) = |x|$



$$f'(x) = 1 \text{ if } x > 0$$

$$f'(x) = -1 \text{ if } x < 0$$

$$f'(x) \text{ undefined if } x = 0$$

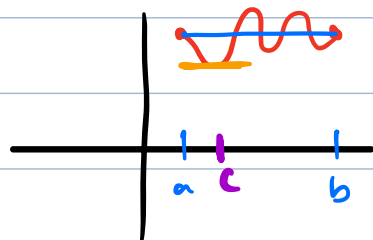
if we tried to apply MVT: on $[-1, 1]$

$$f'(c) \stackrel{?}{=} \frac{f(b) - f(a)}{b - a} = \frac{f(1) - f(-1)}{1 - (-1)} = 0 \quad \times$$

MVT fails.

Before we can prove the MVT, let's take a look at a special case:
what if $f(a) = f(b)$?

then $\frac{f(b) - f(a)}{b - a} = 0$



Rolle's Theorem (special case of MVT): Let f be a function satisfying:

- i. f continuous on $[a, b]$
- ii. f differentiable on (a, b)
- iii. $f(a) = f(b)$

Then there must exist some c such that $a < c < b$ and
 $f'(c) = 0$

Proof idea:



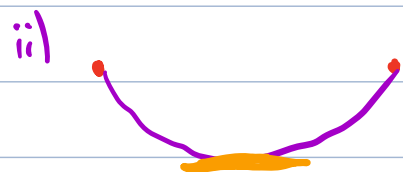
we have 3 cases:

in order to end up where we started
we either

i) increase, then decrease

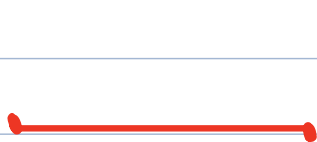
or

ii) decrease, then increase



or

iii) stay constant



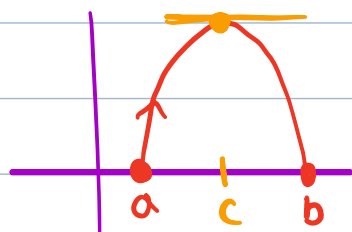
In case i), there must be a local max $\Rightarrow f'(c) = 0$

In case ii) there must be a local min $\Rightarrow f'(c) = 0$

In case iii) $f'(c) = 0$ everywhere.

Applications of Rolle's Theorem

Example: throw a ball up in the air. Then there must be a point where its velocity (in the vertical direction) is 0.



observation: if $\underline{f'(x) > 0}$ for x in $[a, b]$ then $f(a) \neq f(b)$

Explanation using Rolle's theorem:

if $f(a) = f(b)$, then by Rolle's Theorem

there exists some $a < c < b$ where $f'(c) = 0$.

But by assumption $f'(c) > 0$. So this is impossible.

Therefore $f(a) \neq f(b)$

Example: Prove that the equation $\underbrace{x^3 + x - 1}_{f(x)} = 0$ has exactly one real root.
point where $f(x) = 0$

Idea: i) at least one real root

ii) at most one real root

i) show at least one root: use IVT

$$f(x) = 0.$$

idea: $f(2) = 2^3 + 2 - 1 = 8 + 2 - 1 = 9$

$$f(0) = 0^3 + 0 - 1 = -1$$



$$-1 < 0 < 9$$

therefore by IVT (since $f(x)$ is continuous)
there must be some x such that $f(x) = 0$

ii) show at most one root.

Suppose that a and b are both roots.

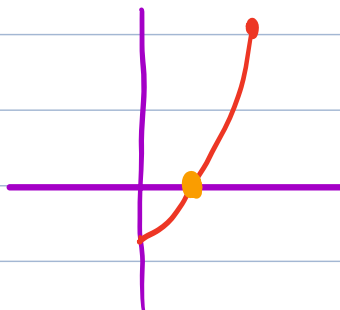
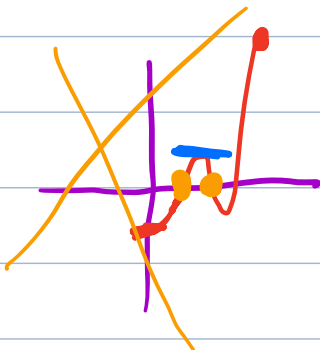
$$f(a) = 0 = f(b)$$

So by Rolle's Theorem (f is continuous and differentiable)

there must be some number c such that $f'(c) = 0$
show this is impossible

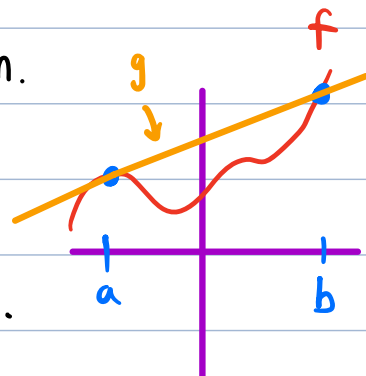
$$f(x) = x^3 + x - 1 \Rightarrow f'(x) = 3x^2 + 1 > 0$$

$\Rightarrow$ cannot have $f'(c) = 0 \Rightarrow$ cannot have two roots.



Back to MVT...

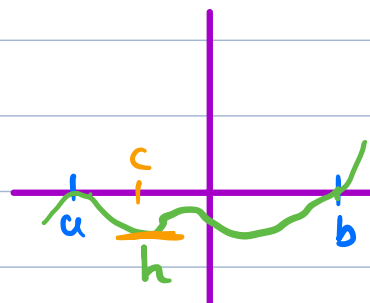
Let's now return to the general mean value Theorem.
We can actually use Rolle's Theorem to prove it!



Idea: Let $y=f(x)$ satisfying conditions of MVT.

Let $y=g(x)$ represent the secant line between
 $(a, f(a))$ and $(b, f(b))$

Note: $g(a)=f(a)$ $g(b)=f(b)$



Let $\underline{h(x) = f(x) - g(x)}$

$$h(a) = 0$$

$$h(b) = 0$$

use Rolle's Thm for h !

So we know $h'(c) = 0$ for some $a < c < b$

$$h'(x) = f'(x) - \underbrace{g'(x)}_{\text{slope of secant line}} = \frac{f(b) - f(a)}{b - a}$$

$$0 = h'(c) = f'(c) - g'(c)$$

$$\boxed{f'(c) = g'(c) = \frac{f(b) - f(a)}{b - a}} \quad \text{MVT}$$

Applications of MVT

Example: Suppose $f(x)$ is a differentiable function.

All we know about it is that $f(0) = -3$ and $f'(x) \leq 5$ for all x .

What is the largest value that $f(2)$ could possibly be?

Solution:

Intuition: Since $f' \leq 5$, f increases at a rate ≤ 5

so when x changes by 2, y changes by $\leq 2 \cdot 5 = 10$

$$\frac{\Delta y}{\Delta x} \leq 5 \Rightarrow \Delta y \leq 5 \Delta x$$

$$f(2) \leq f(0) + 10 = -3 + 10 = 7$$

Formalize using MVT:

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad a=0, b=2$$

$$f'(c) = \frac{f(2) - \overbrace{f(0)}^{-3}}{2-0} \Rightarrow f'(c) = \frac{f(2) + 3}{2}$$

$$f(2) = \underbrace{2f'(c)}_{\leq 5} - 3$$

$$\leq 2 \cdot 5 - 3 = 10 - 3 = \boxed{7}$$

Example Suppose that $3 \leq f'(x) \leq 5$ for all values of x .
Show that

$$18 \leq f(8) - f(2) \leq 30.$$

MVT: $\frac{f(b) - f(a)}{b - a} = f'(c)$ (some c)

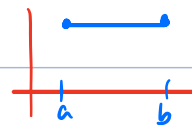
$a=2$ $b=8 \Rightarrow \frac{f(8) - f(2)}{8 - 2} = f'(c)$

$$\frac{f(8) - f(2)}{6} = f'(c) \Rightarrow f(8) - f(2) = 6f'(c)$$

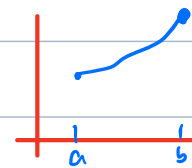
given $3 \leq f'(c) \leq 5 \Rightarrow$ so $\underbrace{6 \cdot 3}_{18} \leq f(8) - f(2) = 6f'(c) \leq \underbrace{6 \cdot 5}_{30}$

Important consequences of MVT:

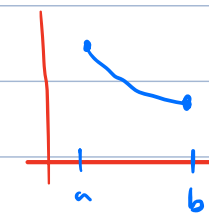
i) if $f'(x) = 0$ for all x , then $f(x)$ is constant ($f(a) = f(b)$)



ii) if $f'(x) > 0$ for all x , then $f(x)$ is increasing ($f(a) < f(b)$)



iii) if $f'(x) < 0$ for all x , then $f(x)$ is decreasing ($f(a) > f(b)$)



Explanations:

for some c

Pick $a < b$. (arbitrary numbers) MVT: $f'(c) = \frac{f(b) - f(a)}{b - a}$

$$i) \quad 0 = \frac{f(b) - f(a)}{b - a} \Rightarrow f(b) - f(a) = 0 \Rightarrow f(a) = f(b) \quad \checkmark$$

$$ii) \quad \frac{f(b) - f(a)}{b - a} > 0 \Rightarrow f(b) - f(a) > 0 \Rightarrow f(b) > f(a) \quad \checkmark$$

$(b - a)$ pos.

$$iii) \quad \frac{f(b) - f(a)}{b - a} < 0 \Rightarrow f(b) - f(a) < 0 \Rightarrow f(b) < f(a) \quad \checkmark$$

$(b - a)$ pos.

Variations:

For all x in some interval

ii') if $f'(x) \geq 0$ then f is nondecreasing over that interval
(if $a < b$, $f(a) \leq f(b)$)

iii') if $f'(x) \leq 0$ then f is nonincreasing
(if $a < b$, $f(a) \geq f(b)$)

Example Use MVT to explain why $e^x > x$ for all $x \geq 0$

Solution:

$$e^x > x \quad \text{when } x \geq 0 \quad \Leftrightarrow \quad \underbrace{e^x - x}_{f(x)} > 0 \quad \text{for all } x \geq 0$$

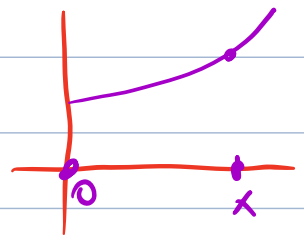
observation: $f'(x) = \underbrace{e^x}_{\geq 1} - 1 \geq 1 - 1 = 0 \quad (\text{if } \underbrace{x \geq 0})$

if $x \geq 0$ by MVT

Since $f'(x) \geq 0$ when $x \geq 0 \Rightarrow f$ is nondecreasing over $[0, \infty)$

$$\Rightarrow f(x) \geq \underbrace{f(0)}_{e^0 - 0 = 1} \quad \text{when } x \geq 0$$

$$\Rightarrow f(x) \geq 1 > 0 \quad \checkmark$$



Example Use MVT to prove the identity $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$

Idea: to show $\tan^{-1} x + \cot^{-1} x = \text{constant}$, want

$$\frac{d}{dx} (\tan^{-1} x + \cot^{-1} x) = 0$$

$$\underbrace{\frac{d}{dx} (\tan^{-1} x)}_{\frac{1}{x^2 + 1}} + \underbrace{\frac{d}{dx} (\cot^{-1} x)}_{-\frac{1}{x^2 + 1}} = 0$$

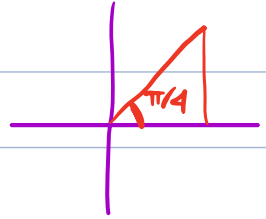
MVT

$$\Rightarrow \tan^{-1} x + \cot^{-1} x = C \quad \text{for some number } C$$

to find C , plug in some value of x .

consider $x=1$: $\tan^{-1}(1) = \pi/4$

$$\cot^{-1}(1) = \pi/4$$



$$\tan^{-1}(1) + \cot^{-1}(1) = \frac{\pi}{4} + \frac{\pi}{4} = \boxed{\frac{\pi}{2}} \Rightarrow C = \frac{\pi}{2}$$

$$\Rightarrow \boxed{\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}}$$